Homework 7 | Matt Kruger

1. Let $\mathsf{NE}_{\mathsf{CFG}} = \{\langle G, H\rangle \mid G$ and H are CFGs with the same set of terminal symbols (that is, alphabet), and $L(G) \neq L(H)\}$. Show that $\mathsf{NE}_{\mathsf{CFG}}$ is Turing-Recognizable. (15 points)

CFG are decidable if it can list its terminals

not sure whether Chomsky is necessary here

"On input $\langle G, H\rangle$:

1. read the CFGs G and H. If their terminal alphabets are NOT IDENTICAL $\rightarrow$ Reject

2. let Σ be the terminal alphabet for CFGs G and H. Enumerate all strings s_i in Σ^*

3. For $i = 1, 2, 3, \ldots$:
   a) run CFG membership decider for both G and H
   b) if only one decider accepts $\rightarrow$ accept
   else continue

4. if all strings exausted (i.e. G and H accept all) $\rightarrow$ reject"

If $L(G) \neq L(H)$ and the terminal sets are equal, there exist some string in Σ^* in the symmetric difference

If $L(G) = L(H)$ M never accepts

Thus, M is a turing machine that recognizes NE_{CFG} and

NE_{CFG} is turing-recognizable

2. A *useless* state in a Turing machine is one that is never entered on any input string. Consider the problem of determining, given a Turing Machine and one of its states, whether the given state is useless. Formulate this problem as a language and show that it is undecidable. (17 points)

$useless_{TM} = \{ \langle M, q \rangle \mid M$ is TM and q is useless state$\}$

Claim: $useless_{TM}$ is undecidable

Proof:

using Reduction from $A_{TM} = \{ \langle M, w \rangle \mid M$ accepts $w \}$

Assume for contradiction that a TM R is a decider for $useless_{TM}$. we use R to construct decider S for A_{TM}

1. M' = "On any input x:

   1. erase the tape
   2. Simulate M on w.
   3. if M enters its accept state q_{accept} then transition to new state q_{new}. Then transition to M' accept state and halt."

2. run R on $\langle M', q_{new} \rangle$:

   - if R accepts $\longrightarrow$ S rejects — Means q_{new} is useless!
   - if R rejects $\longrightarrow$ S accepts — Means q_{new} is not useless!

Case 1: M accepts w $\longrightarrow$ R rejects $\longrightarrow$ S accepts

Case 2: M rejects w $\longrightarrow$ R accepts $\longrightarrow$ S rejects

Thus, S is a decider for A_{TM}, however we know that A_{TM} is undecidable, therefore this is a contradiction

Therefore our initial assumption is incorrect.

$Useless_{TM}$ is undecidable $\square$

3. Let $T = \{\langle M\rangle \mid M$ is a TM that accepts w^R whenever it accepts $w\}$. Note that w^R denotes the reverse of string w. Thus, $\langle M\rangle \in T$ iff TM M has the property that for every input string w, M either accepts both w and w^R or M does not accept w and w^R.

Show that T is undecidable. (18 points)

I.E it halts

Claim: T is undecidable

Proof:

using reduction from $HALT_{TM}$

Assume for a contradiction that TM R is a decider for $HALT_{TM}$
we use R to construct decider S for $HALT_{TM}$

1. M' = "on input x:

   1. if x is not of the form 01 or 10 → reject
   2. if x is 01
      a) simulate M on w.
      b) if simulation halts → accept
   3. if x is 10 → accept "

2. For any string s other than 01 or 10, the reverse property holds, thus since $10 \in L(M')$, $01 \in L(M')$ iff M halts on w.

3. Run R on $\langle M', w\rangle$

   - if R accepts M' → S accepts $\langle M, w\rangle$ $\langle M'\rangle \in T$
   - if R rejects M' → S rejects $\langle M, w\rangle$ $\langle M'\rangle \notin T$

Thus, S is a decider for $HALT_{TM}$, however we know that $HALT_{TM}$ is undecidable, therefore this is a contradiction

Therefore our initial assumption is incorrect.

T is undecidable $\square$